

Fast Fourier transform - the Cooley-Tukey algorithm

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-\frac{2\pi i}{N} nk} = \sum_{n=0}^{N-1} W_{kn} x[n], \quad W_{kn} \equiv e^{-\frac{2\pi i}{N} nk}, \quad k = 0, 1, \dots, N-1$$

Square matrix $\times$ column matrix multiplication = $\mathcal{O}(N^2)$ operations.

The very specific exponential form of W_{kn} depending on the $k \cdot n$ product exhibiting periodicity makes it possible to reduce the number of

operations to

$$\mathcal{O}(N \cdot \log_2 N)$$

$$\begin{pmatrix} \begin{matrix} \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus \\ \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus \\ \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus \\ \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus \\ \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus \\ \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus \\ \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus \\ \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus & \oplus \end{matrix} \end{pmatrix} \cdot \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{pmatrix} = \begin{pmatrix} X_0 \\ X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \\ X_6 \\ X_7 \end{pmatrix}$$

The radix-2 decimation-in-time variant

Separate the terms with even and odd indices of $x[n]$

$$X[k] = \sum_{m=0}^{N/2-1} x[2m]e^{-\frac{2\pi i}{N}(2m)k} + \sum_{m=0}^{N/2-1} x[2m+1]e^{-\frac{2\pi i}{N}(2m+1)k}, \quad k = 0, 1, \dots, N-1$$

In each term the number of outputs is only $N/2$. By limiting k to the $0, 1, \dots, N/2 - 1$ interval we again have the same number of input and output.

$$X[k] = \underbrace{\sum_{m=0}^{N/2-1} x[2m]e^{-\frac{2\pi i}{N/2}mk}}_{DFT(x[2n])} + e^{-\frac{2\pi i}{N}k} \underbrace{\sum_{m=0}^{N/2-1} x[2m+1]e^{-\frac{2\pi i}{N/2}mk}}_{DFT(x[2n+1])} = E[k] + e^{-\frac{2\pi i}{N}k} O[k] \quad \text{for } k = 0, \dots, \frac{N}{2} - 1.$$

$E[k]$ and $O[k]$ are calculated this way for $k = 0, \dots, N/2 - 1$ only and represent DFT-s applied on half of the time points (and returning half of the frequency points).

In order to obtain the other half of the output we use the above formula for $k \geq N/2$ but reindex the output $k \rightarrow k + N/2$ making $k = 0, 1, \dots, N/2 - 1$:

$$X[k + \frac{N}{2}] = \sum_{m=0}^{N/2-1} x[2m]e^{-\frac{2\pi i}{N/2}m(k+\frac{N}{2})} + e^{-\frac{2\pi i}{N}(k+\frac{N}{2})} \sum_{m=0}^{N/2-1} x[2m+1]e^{-\frac{2\pi i}{N/2}m(k+\frac{N}{2})} \quad (1)$$

$$= \sum_{m=0}^{N/2-1} x[2m]e^{-\frac{2\pi i}{N/2}mk} e^{-2\pi mi} + e^{-\frac{2\pi i}{N}k} e^{-\pi i} \sum_{m=0}^{N/2-1} x[2m+1]e^{-\frac{2\pi i}{N/2}mk} e^{-2\pi mi} \quad (2)$$

$$= \sum_{m=0}^{N/2-1} x[2m]e^{-\frac{2\pi i}{N/2}mk} - e^{-\frac{2\pi i}{N}k} \sum_{m=0}^{N/2-1} x[2m+1]e^{-\frac{2\pi i}{N/2}mk} \quad (3)$$

$$= E[k] - e^{-\frac{2\pi i}{N}k} O[k] \quad (4)$$

We can rewrite $X[k]$ as:

$$X[k] = E[k] + e^{-\frac{2\pi i}{N}k} O[k]$$

Cost for an N point DFT is $N^2 \rightarrow$ the cost of the above procedure becomes $2 \cdot \left(\frac{N}{2}\right)^2 + N$.

If $N = 2^P$ we can repeat the halving procedure:

$$2 \cdot \text{DFT} \left(\frac{N}{2} \right) + N \rightarrow 2 \cdot \left(2 \cdot \text{DFT} \left(\frac{N}{4} \right) + \frac{N}{2} \right) + N = 4 \cdot \text{DFT} \left(\frac{N}{4} \right) + 2N$$

$$\rightarrow 4 \cdot \left(2 \cdot \text{DFT} \left(\frac{N}{8} \right) + \frac{N}{4} \right) + 2N = 8 \cdot \text{DFT} \left(\frac{N}{8} \right) + 3N$$

...

$$\rightarrow 2^{P-1} \cdot \left(2 \cdot \text{DFT} \left(\frac{N}{2^P} \right) + \frac{N}{2^{P-1}} \right) + (P-1)N = 2^P \cdot \text{DFT} \left(\frac{N}{2^P} \right) + P \cdot N = N + N \log_2 N$$

```
1 from pylab import *
2 def ditfft2(x):
3     N = len(x)
4     if N == 1:
5         return x
6     else:
7         even = ditfft2(x[::2])
8         odd =
9             ↪ exp(-2*pi*1j/N*arange(N/2))*ditfft2(x[1::2])
10        return hstack([even+odd, even - odd])
```

Fast Fourier transform requires the size of the input vector to be of 2^P .

Exercises

Using the `fft` and `ifft` functions from Python demonstrate the following relations:

1

$$\mathcal{F}\left[e^{-t^2}\right] = \sqrt{\pi}e^{-\pi^2\nu^2}$$

2

$$\mathcal{F}[\text{rect}(t)] = \text{sinc}_{\pi}(f)$$

3

$$\mathcal{F}[\text{sinc}_{\pi}(t)] = \text{rect}(f)$$

Explain the numeric anomaly!

4

$$\mathcal{F}\left[\frac{1}{1+t^2}\right] = \pi e^{-2\pi|\nu|}$$

Use log-normal scale and explain the anomaly!

5

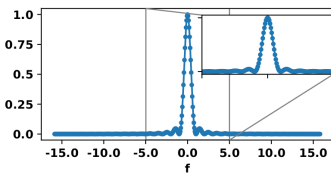
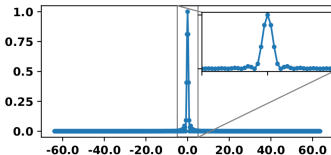
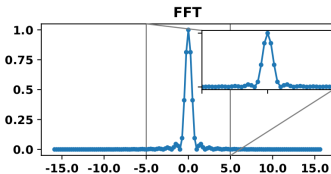
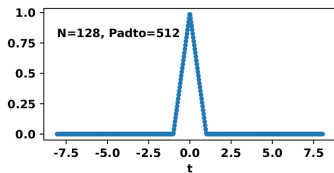
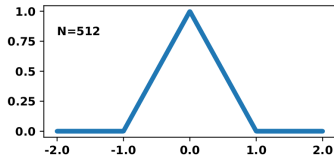
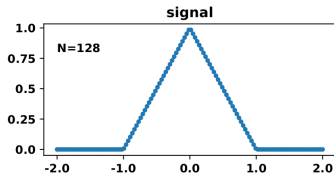
$$\mathcal{F}\left[e^{-|t|}\right] = \frac{2}{1+4\pi^2\nu^2}$$

Use log-log scale, compare the level of numeric anomaly to the previous case and explain it!

Zero padding in FFT

Applied to

- 1 extend sequence lengths to the next higher power of two $2^{\lceil \log_2 N \rceil}$
- 2 smoothen (interpolate) the result



Zero-padding and spectral resolution

Let $x[n]$, $n = 0, \dots, N - 1$, and define the zero-padded signal:

$$x_M[n] = \begin{cases} x[n], & 0 \leq n \leq N - 1 \\ 0, & N \leq n \leq M - 1 \end{cases}$$

DFT as samples of the DTFT:

$$X(e^{j\omega}) = \sum_{n=0}^{N-1} x[n]e^{-j\omega n}$$

$$X_N[k] = X\left(e^{j\frac{2\pi k}{N}}\right), \quad X_M[k] = X\left(e^{j\frac{2\pi k}{M}}\right)$$

If $M = LN$:

$$X_M[kL] = X_N[k]$$

Conclusion:

- $X_M[k]$ samples the **same function** $X(e^{j\omega})$ on a **denser grid**
- original values are preserved; new points are interpolated
- the **spectral shape (main-lobe width)** is unchanged

FFT-based convolution

Continuous case:

$$z(t) \equiv [x * y](t) = \int_{-\infty}^{\infty} x(t')y(t - t') dt'$$

Discrete case:

$$z[n] \equiv (x * y)[n] \equiv \sum_{k=-\infty}^{\infty} x[k]y[n - k] , \quad n \in \mathbb{Z} .$$

Discrete, practical case, $x[0], x[1], \dots, x[p - 1]$ and $y[0], y[1], \dots, y[q - 1]$.

$0 \leq k < p$ and $0 \leq n - k < q$

$$z[n] = \sum_{k=\max\{0, n-(q-1)\}}^{\min\{p-1, n\}} x[k]y[n - k] , \quad n = 0, 1, \dots, p + q - 1$$

$p \cdot q$

number of multiplications.

Exercises

- 1 Show that the number of multiplications for calculating the discrete convolution between two sequences of p respectively q elements is $p \cdot q$.
- 2 Determine the number of additions.

$$z(t) = \mathcal{F}^{-1} [\mathcal{F}[x] \cdot \mathcal{F}[y]] (t)$$

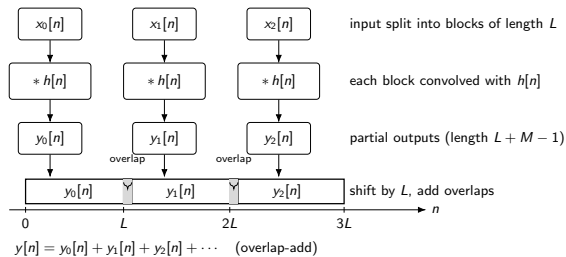
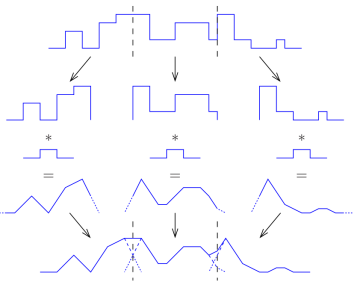
Linear convolution:

$$z[n] = \text{iFFT} [\text{FFT}[x_p] \cdot \text{FFT}[y_p]] [n] , \quad x_p, y_p \text{ zero padded up to } N = p + q - 1.$$

→ $\mathcal{O}(N \log N)$ operations.

Condition: $x[n]$ and $y[n]$ are periodic, with equal period lengths.

- ① In order to prevent a wrap-over both sequences are padded with zero values to a length of at least $p + q - 1$ so that the start and end of the resulting sequence do not overlap.
- ② Thanks to the periodic nature of the DFT, zero padding at both ends has the same effect as padding only at one end
- ③ if one of the sequences are too large or are a realtime waveform that cannot be delayed until the end of the transmission then the sequence has to be split into shorter blocks that are separately convolved and then added together with a suitable overlap.



Deconvolution

$$y(t) = [x * h](t) = \int x(\tau) \cdot h(t - \tau) d\tau \quad |\mathcal{F}[\dots]$$

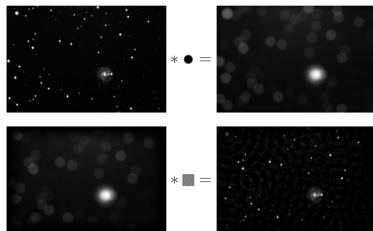
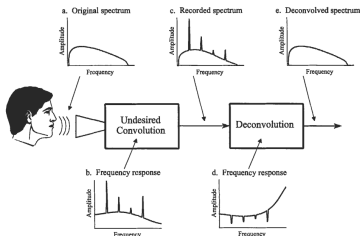
$$Y = X \cdot H \rightarrow X = \frac{Y}{H} \rightarrow x = \mathcal{F}^{-1} \left[\frac{Y}{H} \right]$$

In the presence of noise $s(t)$ (quantization, etc.)

$$y(t) = [x * h](t) + s(t) \quad |\mathcal{F}[\dots]$$

$$Y = X \cdot H + S = \tilde{X} \cdot H \rightarrow \tilde{X} = X + \frac{S}{H} \rightarrow \tilde{x} = x + \mathcal{F}^{-1} \left[\frac{S}{H} \right]$$

Problem: At frequencies where $H(f)$ is close to zero noise will be amplified during deconvolution.

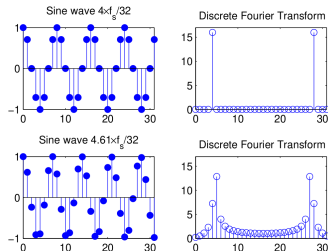


Spectral estimation: an unexpected behavior

Surprises: A sinusoid of period $\tau = 1/f_0$ does not always produce a two-peak spectrum when sampled at $T = 1/f_s$.

1. Sampling should not affect spectrum?

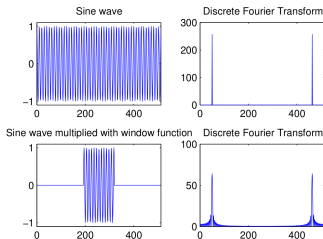
- 1 coherent sampling
 $NT = k\tau \iff f_0 = k \frac{f_s}{N}$
- 2 non-coherent sampling
 $NT \neq k\tau \iff f_0 \neq k \frac{f_s}{N}$



Frequency mismatch

2. Windowing should not affect spectrum?

- Taking only a part of a perfect sinusoid should not change its frequency content?
- Adding zeros should not introduce new frequencies?



Spectral estimation: Leakage and scalloping

Two effects are involved:

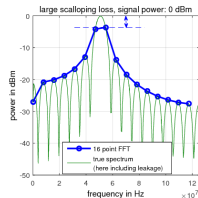
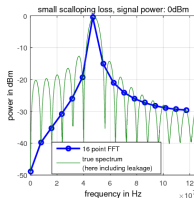
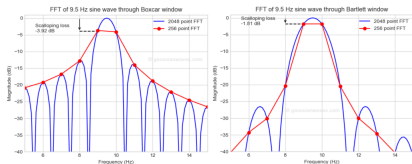
- Windowing: $x[n] \rightarrow x[n] w[n]$ (finite observation)
- Frequency sampling: sampling at $f_k = k \frac{f_s}{N}$

Result:

$$X[k] \approx (X_{\text{true}} * W)(f_k)$$

- Windowing $\rightarrow$ modifies the spectrum by adding an extra temporal scale.
- Frequency sampling $\rightarrow$ important frequencies might be missing

- 1 **Leakage:** energy spreading due to convolution with $W(f)$
- 2 **Scalloping loss:** amplitude underestimation caused by sampling a continuous spectrum at discrete FFT bin frequencies



Revisiting coherent sampling

coherent sampling is just a special case

$$NT = k\tau \iff f_0 = k \frac{f_s}{N}$$

- sampling grid hits the peak of the spectrum
- zeros of $W(f)$ align with other bins

Result:

- no visible leakage
- no scalloping loss
- the periodicity of the windowing has no effect $\rightarrow$ a single time scale of the oscillation

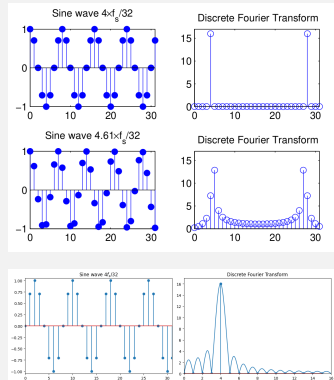
In the general case:

$$f_0 \neq k \frac{f_s}{N}$$

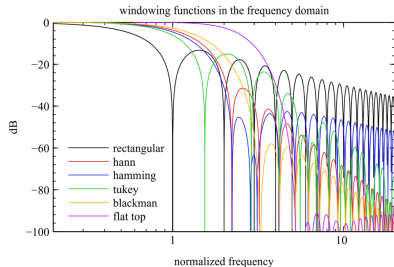
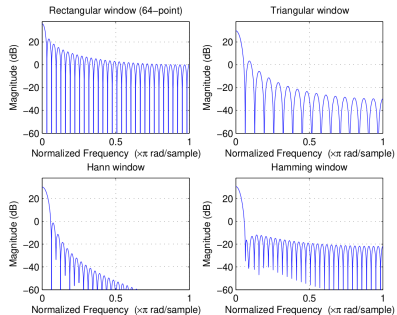
- peak lies between bins
- sampled away from maximum

Key idea:

Coherent sampling does not remove the effects — it aligns them with the DFT grid so they become invisible.



Effect of windowing



Logarithmic units:

$$1 \text{ Np} : \frac{x_1}{x_2} = e \quad 1 \text{ B} = \frac{1}{2} \ln(10) \text{ Np} = 10 \text{ dB}$$

Rectangular (Dirichlet)

$$w[n] = 1$$

Triangular (Bartlett)

$$w[n] = 1 - \left| \frac{n - \frac{N}{2}}{\frac{N}{2}} \right|, \quad 0 \leq n \leq N$$

Hann

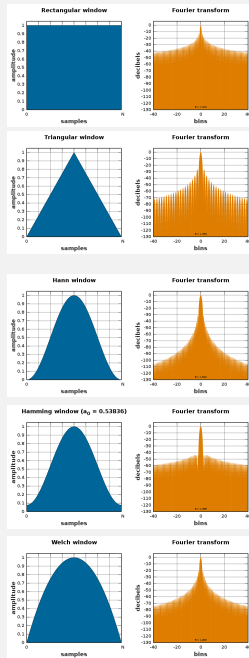
$$w[n] = 0.5 \left[1 - \cos \left(\frac{2\pi n}{N} \right) \right] = \sin^2 \left(\frac{\pi n}{N} \right)$$

Hamming

$$w[n] = 0.54 - 0.46 \cdot \cos \left(\frac{2\pi n}{N} \right)$$

Welch

$$w[n] = 1 - \left(\frac{n - \frac{N}{2}}{\frac{N}{2}} \right)^2, \quad 0 \leq n \leq N$$



$$w[n] = a_0 - a_1 \cos\left(\frac{2\pi n}{N-1}\right) + a_2 \cos\left(\frac{4\pi n}{N-1}\right) - a_3 \cos\left(\frac{6\pi n}{N-1}\right) + a_4 \cos\left(\frac{8\pi n}{N-1}\right), \quad 0 \leq n \leq N-1$$

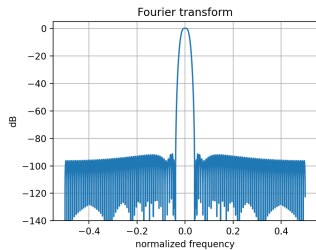
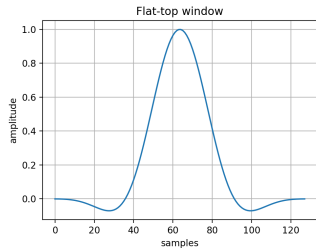
$$a_0 = 0.21557895$$

$$a_1 = 0.41663158$$

$$a_2 = 0.27726316$$

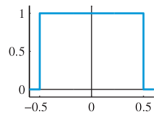
$$a_3 = 0.08357895$$

$$a_4 = 0.00694737$$

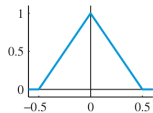


Rectangular

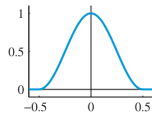
Time-Domain



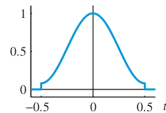
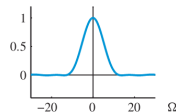
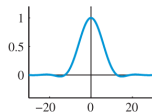
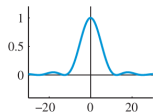
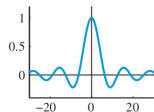
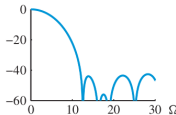
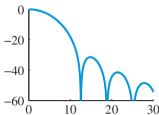
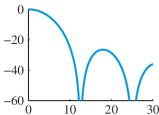
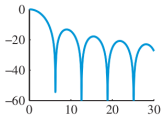
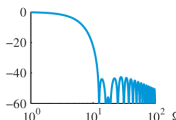
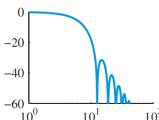
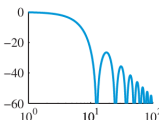
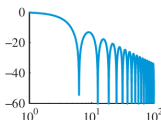
Bartlett



Hann



Hamming

Frequency-Domain
(Linear-Linear)Frequency-Domain
(Log-Linear)Frequency-Domain
(Log-Log)

Window	Mainlobe width	Rolloff rate (dB/octave)	Peak sidelobe level (dB)
Rectangular	$4\pi/T_0$	-6	-13.3
Bartlett	$8\pi/T_0$	-12	-26.5
Hann	$8\pi/T_0$	-18	-31.5
Hamming	$8\pi/T_0$	-6	-42.9

 T_0 - window width